

True integer value time series

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Abstract We construct an integer-valued stationary symmetric AR(1) process which can have either a positive or a negative lag-one autocorrelation. Nearly all integer-valued time series models are designed for observations which are non-negative integers or counts. They have innovations which are distributed on the non-negative integers and therefore obviously non-symmetric. We build our model using innovations that come from the difference of two independent identically distributed Poisson random variables. These innovations have a symmetric distribution, which has many advantages; in particular, they will allow us to model negative correlations. For our AR(1) process, we examine its basic properties and consider estimation via conditional least squares.

Keywords Integer · Positive correlations · Negative correlations · Symmetric innovations

1 Introduction

In this paper, we develop models which are build on the sample space of the integers (including both negative and positive integers). Almost all the integer-valued time series models to date appearing in the literature have innovations which are distributed on the non-negative integers and therefore obviously non-symmetric. The use of innovations that come from a symmetric distribution has many advantages, and allows us to create processes which have more commonalities with the traditional Box–Jenkins ARIMA models. The innovations considered here parallel the Poisson process in construction.

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Poisson ARMA models are constructed with the binomial thinning operator, see Steutel and van Harn (1979). Many properties of these models are found in McKenzie (1988). In particular, Alzaid and Al-Osh (1987) examine the INAR(1) model, covering basic properties and estimation. Inference for this model is developed in Freeland and McCabe (2004a). The issues of forecasting low count times series and in particular forecasting the INAR(1) model are found in Freeland and McCabe (2004b).

The purpose of this paper is to extend discrete time series models over the non-negative integers to models over the integers. More specifically, we extend the Poisson INAR(1) model to a symmetric model around zero. One of the main additional features that this model incorporates is both positive and negative correlation.

The rest of this paper is organized as follows. Section 2 develops the discrete random walk from which we define our innovations, which are integer-valued and symmetric around zero. In Sect. 3, we define our model, which we call the true integer autoregressive model of order 1, or simply TINAR(1). Then in Sect. 4, we outline some simple ways to estimate the model using conditional least squares. We then fit the model to a stock transaction data set, where at each transaction time the change in the stock price is recorded. Finally in the last section, we conclude by summarizing our contributions and suggesting extensions to the model.

2 Discrete random walk

The random walk process which we define in this section will be the building block for the construction of our model. More specifically, the changes in the random walk process will be used as our innovations. This process is a simple extension of the Poisson process to include the negative integers.

According to Ross (1983), a counting process $N(t)$, $t \geq 0$, is a Poisson process if: $N(0) = 0$, the process has stationary and independent increments, $P(N(h) = 1) = \lambda h + o(h)$, and $P(N(h) \geq 2) = o(h)$. We simply extend this to include jumps which are both positive and negative. Thus we have

$$P(R(h) = 1) = P(R(h) = -1) = \lambda h + o(h)$$

and

$$P(R(h) \geq 2) = P(R(h) \leq -2) = o(h)$$

where $R(t)$ is our process. This process is a little different than a counting process, since movement occurs in two directions rather than only increasing. Thus we call this process a random walk with discrete steps of size one.

We define the processes probability function as $P_n(t) = P(R(t) = n)$, $n = 0, \pm 1, \pm 2, \dots$. Since the process is symmetric about zero, we have $P_n(t) = P_{-n}(t)$. Further, we can show

$$\begin{aligned} P_n(t+h) &= P(R(t+h) = n) \\ &= P_n(t)(1 - 2\lambda h) + P_{n-1}(t)\lambda h + P_{n+1}(t)\lambda h + o(h), \end{aligned}$$

which leads to the following differential equation

$$P'_n(t) = \lambda(P_{n+1}(t) + P_{n-1}(t) - 2P_n(t)), \quad (1)$$

and for $n = 0$, this reduces to

$$P'_n(t) = 2\lambda(P_1(t) - P_0(t)). \quad (2)$$

Further, because of symmetry we need only examine non-negative values for n . Another way to define our random walk process is by taking a difference of two independent Poisson processes. That is, let

$$R(t) = N_1(t) - N_2(t),$$

where $N_1(t)$ and $N_2(t)$ are two independent identically distributed (I.I.D.) Poisson processes with rate λ . To see that these are the same, consider the probability function. Firstly,

$$\begin{aligned} P(R(h) = 1) &= P(N_1(h) = 1 \text{ \& } N_2(h) = 0) \\ &= (\lambda h + o(h))(1 - \lambda h + o(h)) = \lambda h + o(h) \end{aligned}$$

and

$$\begin{aligned} P(R(h) = 0) &= P(N_1(h) = 0 \text{ \& } N_2(h) = 0) + P(N_1(h) = N_2(h) \text{ \& } N_1(h) \geq 1) \\ &= (1 - \lambda h + o(h))^2 + (\lambda h + o(h))^2 = 1 - 2\lambda h + o(h). \end{aligned}$$

Finally, $P(R(h) = -1) = P(R(h) = 1)$ due to symmetry, and all other cases are of order $o(h)$. This representation of the process immediately gives us many of its properties. Firstly, consider the moment generating function which is

$$\begin{aligned} E[e^{sR(t)}] &= E[e^{sN_1(t)}]E[e^{-sN_2(t)}] \\ &= e^{\lambda t(e^s + e^{-s} - 2)}. \end{aligned} \quad (3)$$

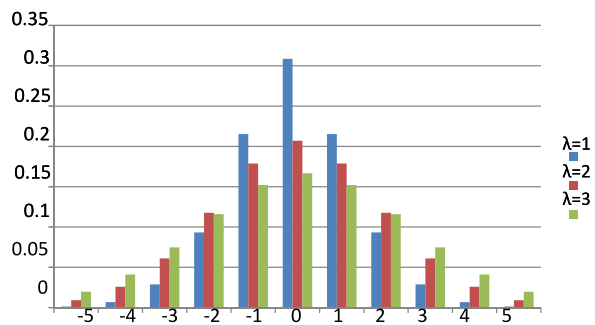
From this the moments are easily calculated. The second and fourth moments are respectively $2\lambda h$ and $2\lambda h + 12\lambda^2 h^2$. All the odd moments are zero because of the symmetry around zero. For a fixed value of t , our process is just the difference of two I.I.D. Poisson random variables with mean λt . This distribution was examined by Irwin (1937). In this paper, Irwin derives the moment generating function, given above in (3), and the probability function which is

$$P_n(t) = e^{-2\lambda t} I_n(2\lambda t), \quad (4)$$

where $I_n(2\lambda)$ is the modified Bessel function of the first kind of order n with argument 2λ . This probability function, of course, satisfies our two differential equations (1) and (2).

Next, we define our innovations as $\varepsilon_t = R(t) - R(t-1)$ for $t = 1, 2, \dots$. These innovations have zero mean and variance 2λ and can thus be parameterized simply by their variance. Further, they have a very simple moment generating function,

Fig. 1 The probability distribution of the innovations for different values of λ



$\exp(\lambda(e^s + e^{-s} - 2))$. A more general version of this distribution is called the Skellam distribution (Skellam 1946), which is the difference between two independent but not identical Poisson random variables. We will refer to the distribution of our innovations as a Skellam distribution with a single parameter λ , which is the mean of the two underlying Poisson random variables. The more general version of the Skellam distribution has two parameters, one for each of the underlying Poisson random variables. In Fig. 1, we depict the probability distribution of our innovations for $\lambda = 1, 3$ and 5.

Joe (1996) developed a method to create stationary time series from any innovation process within the class of convolution-closed infinitely divisible distributions. This means that if the innovations were added together they would remain in the same distribution. Further, for any innovation we can find a sum of I.I.D. random variables such that their sum is equal to the innovation in distribution; for more on infinitely divisibility, see (Steutel 1979). These are both very attractive features that Gaussian innovations also share, and, in fact, Gaussian innovations are included in Joe's method.

Our new innovations extend the list of attractive features even further. Not only are these innovations convolution-closed infinitely divisible, but they are also symmetric and can also be subtracted from one another while still remaining in the same distribution. This makes them even more similar to Gaussian innovations. Thus for $t \neq s$, if ε_t and ε_s are I.I.D. $\text{Skellam}(\lambda)$ then $\varepsilon_t + \varepsilon_s$ and $\varepsilon_t - \varepsilon_s$ are both $\text{Skellam}(2\lambda)$. To see this, let

$$\varepsilon_t = \epsilon_t - \eta_t, \quad (5)$$

where ϵ_t and η_t are independent Poisson random variables each with mean λ . Then we have for $t \neq s$, $\varepsilon_t - \varepsilon_s = (\epsilon_t + \eta_s) - (\eta_t + \epsilon_s)$, where $\epsilon_t + \eta_s$ and $\eta_t + \epsilon_s$ are independent $\text{Poisson}(2\lambda)$. We can think of the random variables ϵ_t and η_t as forces in opposite directions. Thus if $\epsilon_t > \eta_t$ then we observe $\varepsilon_t > 0$, when $\epsilon_t < \eta_t$ we observe a negative integer, and when they are equal we observe zero.

3 TINAR(1)

In this section, we will define the true integer-valued autoregressive process of order one. The innovations will come from our random walk process, which is symmetri-

cally distributed about zero over the integers. One advantage of this is that we can allow negative correlations.

We begin by defining the thinning operation. First, let $Z_{t-1} \sim \text{Skellam}(\frac{\lambda}{1-\alpha})$ which, of course, can be written as the difference of two Poisson random variables. Thus, let $Z_{t-1} = X_{t-1} - Y_{t-1}$, where X_{t-1} and Y_{t-1} are latent independent Poisson random variables with mean $\frac{\lambda}{1-\alpha}$. When X_{t-1} is larger than Y_{t-1} then our observation Z_{t-1} is positive, and conversely, if Y_{t-1} is larger than X_{t-1} then our observation is negative. Our new thinning operator, $\star$, is binomial thinning acting on the two latent random variables. We denote the binomial thinning operator as $\circ$, which is defined for a non-negative integer M and $0 \leq \alpha < 1$ as $\alpha \circ M = \sum_{k=1}^M B_k$, where B_k are independent identically distributed Bernoulli trials with probability of success α . Our new operator, $\star$, is defined in the following way for positive correlations

$$\alpha \star Z_{t-1} | Z_{t-1} = \alpha \circ X_{t-1} - \alpha \circ Y_{t-1} | X_{t-1} - Y_{t-1}, \quad (6)$$

for $0 \leq \alpha < 1$.

We next explore the conditional distribution of $\alpha \star Z_{t-1}$, which we do in three parts. First conditioning on the latent process, we have

$$\begin{aligned} P(\alpha \star Z_{t-1} = w | X_{t-1} = z + y, Y_{t-1} = y) \\ = \sum_{l=0}^y \binom{z+y}{w+l} \alpha^{w+l} (1-\alpha)^{z+y-w-l} \binom{y}{l} \alpha^l (1-\alpha)^{y-l}, \end{aligned} \quad (7)$$

where $\binom{a}{b} = 0$ if $b \leq 0$ or $b > a$. Next the joint distribution of X_{t-1} and Y_{t-1} is

$$P(X_{t-1} = z + y, Y_{t-1} = y) = \frac{e^{-\frac{2\lambda}{1-\alpha}} \left(\frac{\lambda}{1-\alpha}\right)^{z+2y}}{(z+y)!y!}, \quad (8)$$

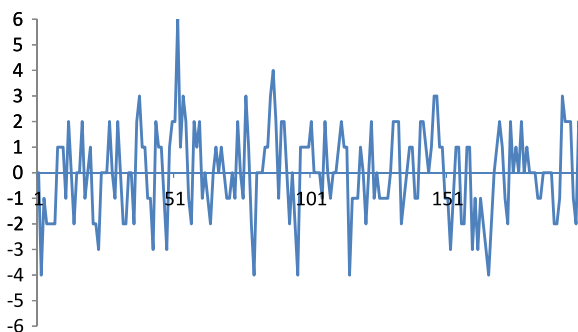
where $x! = 0$ for $x \leq 0$. Then finally, the conditional distribution of $\alpha \star Z_{t-1}$ given Z_{t-1} is found by taking the product of (7) and (8) and then summing over y . Without loss of generality, let $z > 0$ and let w be unrestricted. Then

$$P(\alpha \circ Z_{t-1} = w | Z_{t-1} = z) = \sum_{y=0}^{\infty} \frac{e^{-\frac{2\lambda}{1-\alpha}} \lambda^{z+2y}}{(z+y)!y!} \sum_{l=0}^y \binom{z+y}{w+l} \binom{y}{l} \left(\frac{\lambda}{1-\alpha}\right)^{w+2l}, \quad (9)$$

and because of symmetry $P(\alpha \circ Z_{t-1} = w | Z_{t-1} = -z) = P(\alpha \circ Z_{t-1} = -w | Z_{t-1} = z)$. While this expression is somewhat messy, we are able to find many of the properties of our process without directly using this conditional probability function.

Suppose that time series $\{Z_t : t = 0, \pm 1, \pm 2, \dots\}$ is integer-valued and with a Skellam marginal distribution with a single parameter $\frac{\lambda}{1-\alpha}$. Also suppose that ε_t 's are independent and identically distributed with a Skellam distribution with a single parameter λ .

Fig. 2 A sample path of length 200 with $\lambda = 1$ and $\alpha = 0.3$



Definition 1 Let $\{Z_t\}$ be an integer-valued stochastic process such that

$$Z_t = \alpha \star Z_{t-1} + \varepsilon_t \quad (10)$$

where the $\star$ operator is as defined in (6).

We call this process the True Integer-valued Autoregressive model of order 1, or TINAR(1). It is illuminating to view the TINAR(1) process as the difference between two independent Poisson INAR(1) processes. Let $X_t = \alpha \circ X_{t-1} + \varepsilon_t$ and $Y_t = \alpha \circ Y_{t-1} + \eta_t$, where ε_t and η_t , $t = 1, 2, \dots$, are independent Poisson(λ) and X_0 and Y_0 are independent Poisson($\frac{\lambda}{1-\alpha}$), “ $\circ$ ” is the usual binomial thinning operator and $0 < \alpha < 1$. For positive correlation, we let $Z_{t-1} = X_{t-1} - Y_{t-1}$, then we have

$$\begin{aligned} Z_t &= \alpha \star Z_{t-1} + \varepsilon_t \\ &= \alpha \circ X_{t-1} + \varepsilon_t - \alpha \circ Y_{t-1} - \eta_t \\ &= X_t - Y_t. \end{aligned}$$

At the end of this section we will show how negative correlations can be modeled, but to simplify the presentation of the following results we will only consider positive correlations. A sample path for a generated series of length 200 with parameters $\lambda = 1$ and $\alpha = 0.3$ is given in Fig. 2.

Because the TINAR(1) process can be viewed as the difference between two independent Poisson INAR(1) processes, this gives us many properties for our process. The following five properties appear in Alzaid and Al-Osh (1987):

$$\begin{aligned} X_t &\stackrel{d}{=} \sum_{j=0}^{\infty} \alpha^j \circ \varepsilon_{t-j}, \\ (X_t, X_{t-k}) &\stackrel{d}{=} \left(\alpha^k \circ X_{t-k} + \sum_{j=0}^{k-1} \alpha^j \circ \varepsilon_{t-j}, X_{t-k} \right), \\ E[X_t] &= \alpha E[X_{t-1}] + \mu = \alpha^t E[X_0] + \mu \sum_{j=0}^{t-1} \alpha^j, \end{aligned}$$

$$\text{var}(X_t) = \alpha^{2t} \text{var}(X_0) + (1 - \alpha) \sum_{j=1}^t \alpha^{2j-1} E[X_{t-j}] + \sigma^2 \sum_{j=1}^t \alpha^{2(j-1)},$$

and

$$\gamma(k) = \text{cov}(X_{t-k}, X_t) = \alpha^k \text{var}(X_{t-k}) = \alpha^k \gamma(0),$$

where $\mu = E[\epsilon_t]$ and $\sigma^2 = \text{var}(\epsilon_t)$. For the TINAR(1) process, we have the following corresponding formulas:

$$Z_t = X_t - Y_t \stackrel{d}{=} \sum_{j=0}^{\infty} \alpha^j \circ \epsilon_{t-j} - \sum_{j=0}^{\infty} \alpha^j \circ \eta_{t-j} = \sum_{j=0}^{\infty} \alpha^j \star \varepsilon_{t-j}, \quad (11)$$

$$(Z_t, Z_{t-k}) \stackrel{d}{=} \left(\alpha^k \star Z_{t-k} + \sum_{j=0}^{k-1} \alpha^j \star \varepsilon_{t-j}, Z_{t-k} \right), \quad (12)$$

$$E[Z_t] = E[X_t] - E[Y_t] = \alpha E[Z_{t-1}] = \alpha^t E[Z_0], \quad (13)$$

$$\text{var}(Z_t) = \text{var}(X_t) + \text{var}(Y_t)$$

$$\begin{aligned} &= \alpha^{2t} \text{var}(Z_0) + (1 - \alpha) \sum_{j=1}^t \alpha^{2j-1} (E[X_{t-j}] + E[Y_{t-j}]) \\ &\quad + 2\sigma^2 \sum_{j=1}^t \alpha^{2(j-1)}, \end{aligned} \quad (14)$$

and

$$\begin{aligned} \gamma(k) &= \text{cov}(Z_{t-k}, Z_t) \\ &= \text{cov}(X_{t-k}, X_t) + \text{cov}(Y_{t-k}, Y_t) \\ &= \alpha^k (\text{var}(X_{t-k}) + \text{var}(Y_{t-k})) \\ &= \alpha^k \text{var}(Z_{t-k}) \\ &= \alpha^k \gamma(0). \end{aligned}$$

For the INAR(1) model, Alzaid and Al-Osh (1987) give the second order stationarity conditions $E[X_0] = \frac{\mu}{1-\alpha}$ and $\text{var}(X_0) = (\alpha\mu + \sigma^2)/(1 - \alpha^2)$. Further, if ϵ_t is Poisson(λ) then the process is strongly stationary and the stationary distribution of X_t is Poisson($\frac{\lambda}{1-\alpha}$). Under this condition, the TINAR(1) process is also strongly stationary and the stationary distribution of Z_t is Skellam($\frac{\lambda}{1-\alpha}$). In Jin-Guan and Yuan (1991), the INAR(p) model is shown to be ergodic under certain conditions, which are satisfied in the INAR(1) model for $0 < \alpha < 1$. Further, Freeland (1998) showed that the INAR(1) process is α -mixing. Since the TINAR(1) model is the difference between two independent INAR(1) processes, it must also be ergodic and α -mixing.

For negative correlations, we can again let $Z_{t-1} = X_{t-1} - Y_{t-1}$ and $0 < \alpha < 1$, but define the process as

$$\begin{aligned} Z_t &= \alpha \star -Z_{t-1} + \varepsilon_t \\ &= -\alpha \circ X_{t-1} - \epsilon_t + \alpha \circ Y_{t-1} + \eta_t \\ &= Y_t - X_t. \end{aligned}$$

Thus the differences in the latent processes are alternating or we could write

$$Z_t = \begin{cases} X_t - Y_t & \text{if } t = 0, 2, 4, \dots, \\ Y_t - X_t & \text{if } t = 1, 3, 5, \dots \end{cases}$$

Here we have defined $\varepsilon_t = \epsilon_t - \eta_t$ for $t = 0, 2, 4, \dots$ and $\varepsilon_t = \eta_t - \epsilon_t$ for $t = 1, 3, \dots$, which in either case has a Skellam(λ) distribution. For negative correlations, Formulas (11)–(13) and (15) can be rewritten as

$$Z_t \stackrel{d}{=} \sum_{j=0}^{\infty} \alpha^j \star (-1)^j \varepsilon_{t-j}, \quad (15)$$

$$(Z_t, Z_{t-k}) \stackrel{d}{=} \left(\alpha^k \star (-1)^k Z_{t-k} + \sum_{j=0}^{k-1} \alpha^j \star (-1)^j \varepsilon_{t-j}, Z_{t-k} \right), \quad (16)$$

$$E[Z_t] = E[X_t] - E[Y_t] = -\alpha E[Z_{t-1}] = (-\alpha)^t E[Z_0], \quad (17)$$

and

$$\gamma(k) = \text{cov}(Z_{t-k}, Z_t) = (-\alpha)^k \gamma(0). \quad (18)$$

4 Estimation

Since the likelihood for this model is cumbersome to work with, we will use Yule–Walker and conditional least squares to estimate the model parameters. Since the conditional mean is αZ_{t-1} , conditional least squares will yield only an estimate for α and not for λ . Using conditional least squares results in the following estimating equation for α

$$\sum_{t=1}^n (Z_t - \alpha Z_{t-1}) Z_{t-1}$$

and the following estimate

$$\hat{\alpha}_n = \frac{\sum_{t=1}^n Z_{t-1} Z_t}{\sum_{t=1}^n Z_{t-1}^2}. \quad (19)$$

The Yule–Walker estimate is asymptotically equivalent and is given by

$$\hat{\alpha}_n = \frac{\sum_{t=1}^n Z_{t-1} Z_t}{\sum_{t=1}^{n+1} Z_{t-1}^2}. \quad (20)$$

Next we consider the asymptotic distribution of this estimate.

Theorem 1 *If the process Z_t is defined as in Definition 1, then the conditional least squares estimate for α has the following asymptotic distribution*

$$n^{\frac{1}{2}}(\hat{\alpha}_n - \alpha) \xrightarrow{d} N(0, \sigma^2),$$

where

$$\sigma^2 = 2\lambda + 2\alpha(1 - \alpha) + 4\alpha\lambda$$

and $\hat{\alpha}_n$ is as given in (19).

This result is rather straightforward to prove since our process is stationary ergodic with finite moments.

Proof First, note that

$$n^{\frac{1}{2}}(\hat{\alpha}_n - \alpha) = \frac{n^{-\frac{1}{2}} \sum_{t=1}^n Z_{t-1}(Z_t - \alpha Z_{t-1})}{n^{-1} \sum_{t=1}^n Z_{t-1}^2},$$

and further that the denominator converges almost surely to $\frac{2\lambda}{1-\alpha}$. Therefore, by the law of large numbers and the continuous mapping theorem, this has the same asymptotic distribution as

$$\frac{1-\alpha}{2\lambda} n^{-\frac{1}{2}} \sum_{t=1}^n Z_{t-1}(Z_t - \alpha Z_{t-1}).$$

Since this sum is a martingale and the martingale differences are stationary ergodic with finite moments, it converges to a normal distribution with mean zero and variance $\frac{1-\alpha}{2\lambda} E[Z_{t-1}^2(Z_t - \alpha Z_{t-1})^2]$ where the expected value can be calculated as

$$\begin{aligned} E[Z_{t-1}^2(Z_t - \alpha Z_{t-1})^2] &= E[E[Z_{t-1}^2(Z_t - \alpha Z_{t-1})^2 \mid X_{t-1}, Y_{t-1}]] \\ &= E[(\alpha(1-\alpha)(X_{t-1} + Y_{t-1}) + 2\lambda)(X_{t-1} - Y_{t-1})^2] \\ &= 4\alpha(1-\alpha)(E[X_{t-1}^3] - E[X_{t-1}^2]E[X_{t-1}]) + 4\lambda \text{var}(X_{t-1}) \\ &= \frac{2\lambda}{1-\alpha} [2\lambda + 2\alpha(1-\alpha) + 4\alpha\lambda], \end{aligned}$$

which completes the proof. $\square$

Since the marginal variance is $\frac{2\lambda}{1-\alpha}$, the parameter λ can be estimated by the sample variance as follows:

$$\hat{\lambda}_n = \frac{1 - \hat{\alpha}}{2n} \sum_{t=1}^n Z_t^2. \quad (21)$$

Further, since our process Z_t is α -mixing and stationary ergodic with finite moments, then $n^{\frac{1}{2}}(\hat{\lambda}_n - \lambda)$ either converges in probability to zero or in distribution to a normal distribution with mean zero and some finite variance. The asymptotic variance in this case is more complicated to determine analytically and won't be pursued here. However, in the next section we will use a Monte Carlo simulation to determine standard errors for our estimates.

5 Application to price change data

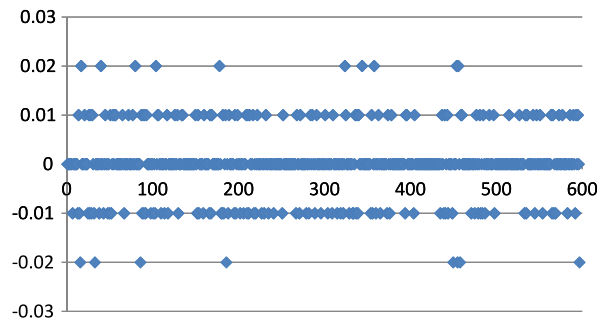
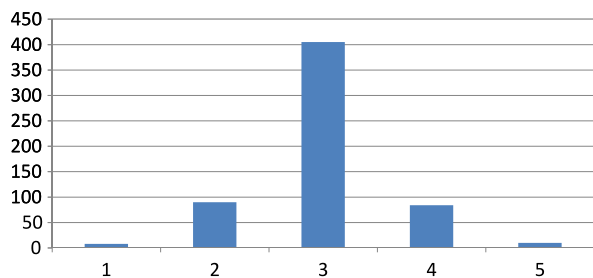
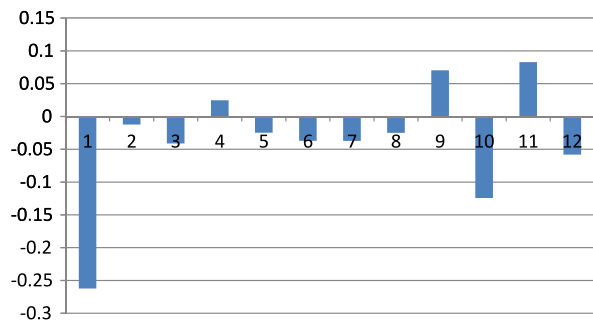
In finance and economics, it is common to model the change in price of a commodity, stock or other financial product. The price of a stock is observed every time that stock is traded, which may happen several hundred times per day. Engle and Russell (1998) use an autoregressive conditional multinomial model to model the price change in each transaction of IBM stock. Their model has 5 states, which are: no price change, up one tick, down one tick, up two or more ticks, and down two or more tick. In this section, we fit our model to price change data for the Australian firm Broken Hill Proprietary (BHP) Limited. BHP was incorporated in 1885 and operated in the mining of silver and lead at Broken Hill, which is in western New South Wales, Australia. In 2001, BHP merged with the British–Dutch company Billiton to form BHP Billiton, which is the world's largest mining company.

Our data consists of the price change, measured in cents, at each transaction time during the day of October 1, 2001. There were 598 transactions resulting in 597 price changes. The time between each transaction varies from 1 second to just over 8 minutes. The median time between transactions is 17 seconds, and the mean time is 34 seconds. Trading each day runs from 10:20 am until 4:00 pm. On this particular day, there does not appear to be much relationship between the duration between transactions and the change in price.

The trade volume per transaction varies from 1 share to 220,000 shares, with a median volume of 1694 shares. Out of 598 trades, 541 of them had trade volumes below 10,000 and 2 had volumes over 50,000. Neither of the two trades with volumes over 50,000 resulted in a price change.

In Figs. 3, 4, and 5, we have respectively the time series plot, histogram and sample correlogram of the price change data. For the time series plot, the time points represent the time ordering of the trades and do not reflect the differing durations between trades. The histogram of the data is consistent with a Skellam distribution with a single parameter. From the correlogram, we see that an AR(1) model with a negative correlation may be suitable for this data set.

We proceeded by fitting our AR(1) model without any covariates as neither duration nor volume appeared to have much influence on this day's price change data.

Fig. 3 Time series plot of the data**Fig. 4** Histogram of the data**Fig. 5** Sample autocorrelation function

We estimated the model parameters using (19) and (21). Then using these estimated parameter values, we simulated 1000 series of length 597 from which we determined the standard errors of the estimation method. This resulted in $\hat{\alpha} = -0.2645$ and $\hat{\lambda} = 0.1515$ with standard errors of 0.04835 and 0.0151, respectively. The simulation showed a small bias in both of the estimates with average values of the estimates of -0.2622 and 0.1517 for α and λ , respectively. Using Theorem 1, we estimated the asymptotic variance to be $\sigma^2 = 1.0232$, which corresponds to a standard deviation for α of $\sigma/\sqrt{597} = 0.041434$, which, as expected, is a little smaller than the standard error found via the simulation.

6 Conclusions

In this paper, we have constructed an intuitive discrete random walk process with unit step sizes, which is similar in construction to the Poisson process. From this process, we construct symmetric integer-valued innovations, which we refer to as having a Skellam distribution. These innovations share many of the properties of Gaussian innovations, in that they are infinitely divisible convolution-closed. Further, they share the property that they may also be subtracted from one another and remain within the same distribution.

We showed how to modify the binomial thinning operator to produce a stationary AR(1) model with a Skellam marginal distribution. This thinning operator is somewhat cumbersome to work with because it is defined on two latent counts for which only the difference is observed. However, even with this restriction many properties were derived; these include the mean, variance and autocorrelations. A nice feature of this model is that the range of the autocorrelation is from -1 to 1 .

Estimation of the model's correlation coefficient is accomplished by conditional least squares and the estimator's asymptotic variance is found. The sample variance was used to estimate λ .

We fit our model to some transaction based price change data. We found the price changes to be negatively correlated and found the histogram of the data to be consistent with the Skellam marginal distribution assumption.

Our TINAR(1) model can be thought of as a difference between two independent Poisson INAR(1) processes. This is different from the work of Weiß (2008), where differences are taken of a single Poisson INAR(1) process at different time points to create a jump process. Given this view of our model, it is quite straightforward to define additional models which have a Skellam marginal distribution, by simply taking the difference between two identical but independent Poisson time series processes. This would include the MA(q), ARMA($1, q$), ARMA($p, p - 1$) and vector AR(1) models considered by McKenzie (1988). The limitation of these models is that the thinning is still defined in terms of a latent process. However, many of the moments should be easy to work out, making moment estimation straightforward.

There are many other possible extensions to our model, here we will suggest three. The simplest would be the addition of an integer mean parameter. We could also make the innovations non-symmetric to handle skewed data. In this case, the innovations could be the difference of two independent Poisson random variables with different means, which is a more general version of the Skellam distribution. The innovations could be based on the difference between two independent identically distributed random variables which are not Poisson; the distributions used could be any in the convolution-closed infinitely divisible class.

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